Tokenizer Fertility Is a Weak Predictor of Zero-Shot Cross-Lingual Transfer

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Abstract

Multilingual encoders are fine-tuned on English and applied zero-shot to other languages, and the loss they suffer varies widely across languages. A common intuition is that this loss comes from the shared subword vocabulary fragmenting some languages more than others. We test a specific version of this claim: that a language's tokenizer fertility relative to English explains 40–60% of the variance in transfer drop, beats typological distance, causes catastrophic failure above 2.5×, and does so independently of pretraining data size. We fine-tune MBERT and XLM-R on English for three tasks (XNLI, WIKIANN NER, SIB-200) and evaluate on up to 71 target languages. The hypothesis is not supported. Fertility explains 0–27% of the between-language variance, below 40% in all six model–task cells, whereas syntactic distance explains 28–68% and exceeds fertility in every cell. Pooled over cells, fertility reaches $R^2 = 0.16$ against 0.37 for syntactic distance and 0.32 for log pretraining size, and it adds at most 0.05 of R^2 once those two are controlled. The 2.5× threshold is not a usable rule: no whitespace-segmented language reaches it under XLM-R's tokenizer. Fragmentation is not harmless, however. Within a language, the model that fragments it more loses more ($r = 0.34$–0.61), and artificially over-segmenting English at test time costs 30–59 points at about twice the normal fertility. Fertility is therefore a weak screening signal for transfer risk, and syntactic distance and pretraining volume are more informative.

1 Introduction

If a multilingual model fails on a language because its tokenizer shreds that language into tiny pieces, we can fix the tokenizer. If it fails because the language is structurally far from English, we cannot. Which of the two explains zero-shot transfer loss therefore decides where practitioners should spend effort, and which languages they should flag as risky before any labeled data exists.

Multilingual encoders such as MBERT Devlin et al. [2019] and XLM-R Conneau et al. [2020] are routinely fine-tuned on English and applied to other languages. The size of the resulting drop differs greatly by language Pires et al. [2019], Wu and Dredze [2020], Hu et al. [2020]. One appealing account points to *tokenizer fertility*, the number of subword tokens per word Ács [2019]. A shared vocabulary is dominated by high-resource languages, so other languages are split into more and shorter pieces, and over-segmentation hurts monolingual downstream quality Rust et al. [2021]. The competing account points to typological distance and pretraining corpus size Pires et al. [2019], Lauscher et al. [2020].

How much of the transfer drop does fertility explain? Existing work studies these factors separately. What is missing is a head-to-head comparison in which fertility, typological distance and pretraining size compete to predict the *same* transfer drops, across several tasks and two tokenizers, with a check of whether fertility is more than a proxy for the other two. We test a concrete hypothesis with four parts: (H1) the fertility ratio relative to English explains 40–60% of the variance in transfer degradation; (H2) it outperforms typological distance; (H3) above 2.5× English fertility,

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transfer fails catastrophically (a drop of more than 30 points); and (H4) the effect is independent of pretraining data size.

We fine-tune MBERT and XLM-R on English for natural language inference (XNLI), named entity recognition (WIKIANN) and topic classification (SIB-200), and evaluate zero-shot on up to 71 target languages that both models saw in pretraining. We fixed the analysis plan before training any model. Because fertility, typology, size and script are correlated across languages, we add two designs that hold the language fixed: a within-language contrast between the two tokenizers, and an intervention that re-segments English test inputs into finer pieces.

The hypothesis is not supported as stated (see figure 1). Fertility explains 0–27% of the between-language variance in transfer drop, below the hypothesized range in all six model–task cells. Syntactic distance from English explains 28–68% and exceeds fertility in every cell; log pretraining size explains clearly more than fertility in four of six. Pooled over cells, fertility reaches $R^2 = 0.16$ [0.07, 0.28], against 0.37 for syntactic distance and 0.32 for log size. Once size and syntactic distance are controlled, fertility adds at most 0.05 of R^2. Yet fragmentation does matter in a narrower sense. For a given language, the model whose tokenizer fragments it more loses more ($r = 0.34$–0.61), and over-segmenting English at test time costs 10–24 points at about $1.4\times$ fertility and 30–59 points at about $2\times$. Fragmentation can hurt; it is just not what separates languages that transfer well from those that do not.

In summary, our contributions are:

- We compare fertility, URIEL typological distances and pretraining size as predictors of the same zero-shot transfer drops over 71 target languages, three tasks and two models, with bootstrap confidence intervals.
- We test the specific numerical claims of the fertility hypothesis and find that none holds in general: R^2 is 0–27%, syntactic distance wins in every cell, and the $2.5\times$ threshold is consistent with the data in one cell of six.
- We separate fertility from language identity with a within-language tokenizer contrast and an over-segmentation intervention, and show that fragmentation has a real cost when it is unfamiliar to the model.
- We report an exploratory lead: a tokens-per-parallel-sentence measure reaches $R^2 = 0.42$–0.51 on SIB-200 and survives controls there, but not on NER or XNLI.

We review related work in section 2, describe the setup in section 3, present results in section 4, and discuss interpretation and limitations in section 5.

2 Related Work

Tokenization quality in multilingual models. Ács [2019] introduced fertility as a way to describe how MBERT's WordPiece vocabulary splits words in different languages. Rust et al. [2021] showed that MBERT over-segments some languages relative to dedicated monolingual tokenizers, and that this hurts monolingual downstream performance. These studies establish that segmentation granularity can matter for a language's quality within a model. Unlike them, we ask whether fertility predicts the *cross-lingual transfer drop* from English, and whether it does so better than competing explanations.

What predicts zero-shot transfer. Pires et al. [2019] related MBERT's zero-shot transfer to typological similarity between source and target, and Wu and Dredze [2019] documented its strong but uneven transfer across tasks. K et al. [2020] studied which linguistic and architectural properties cross-lingual ability depends on. Lauscher et al. [2020] found that transfer performance correlates with both typological similarity and the size of the target-language pretraining corpus, with the mix depending on the task. Wu and Dredze [2020] showed that MBERT serves low-resource languages poorly. Lin et al. [2019] used URIEL distances Littell et al. [2017] together with data-dependent features to rank transfer languages. Building on this line of work, we add fertility as a competing predictor of the same drops, and our results agree with the typology-and-size account rather than replacing it.

Benchmarks. We rely on XNLI Conneau et al. [2018], WIKIANN Pan et al. [2017] with the splits of Rahimi et al. [2019], and SIB-200 Adelani et al. [2024], which is built on FLORES-200

Task	Dataset	English train	Test per language	Targets	Metric
NLI	XNLI	100k subset of MNLI	5,010	14	accuracy
NER	WIKIANN	20,000	100 / 1,000 / 10,000	71	span F1
Topic classification	SIB-200	701	204	71	accuracy

Table 1: Tasks. We train on English only and test zero-shot on every target language.

sentences NLLB Team [2022]. XTREME Hu et al. [2020] popularized the English-train, zero-shot-test protocol that we follow. The wide language coverage of WIKIANN and SIB-200 is what makes a regression over languages feasible.

3 Methodology

3.1 Problem Setup and Hypotheses

For a model m and a task t, we fine-tune on English and measure the test score $s_{m,t}(\ell)$ on each target language ℓ. The dependent variable is the *transfer drop*

$$\Delta_{m,t}(\ell) = s_{m,t}(\text{en}) - s_{m,t}(\ell), \tag{1}$$

in points, averaged over seeds. Relative drop is a constant multiple of Δ within a model–task cell, so R^2 is identical. The language is the unit of analysis within each of the six model–task cells.

We decompose the hypothesis into four pre-specified tests. **H1**: R^2 of Δ regressed on the fertility ratio lies in $[0.40, 0.60]$. **H2**: R^2 of fertility exceeds R^2 of typological distance. **H3**: languages with fertility ratio above 2.5 have $\Delta > 30$, and languages below mostly do not. **H4**: the fertility effect survives controlling for pretraining data size.

3.2 Models, Tasks and Languages

Models. We use MBERT (`bert-base-multilingual-cased`; WordPiece, 119k vocabulary; pretrained on Wikipedia in 104 languages) and XLM-R (`xlm-roberta-base`; SentencePiece unigram, 250k vocabulary; pretrained on CC-100 in 100 languages). We choose these two because they are the standard zero-shot transfer baselines and they have different tokenizers, so a result is not a property of one vocabulary.

Tasks. Table 1 lists the three tasks. We use three so that the result is not a property of one benchmark: XNLI Conneau et al. [2018] trained on MNLI Williams et al. [2018], WIKIANN NER Pan et al. [2017], Rahimi et al. [2019], and SIB-200 topic classification Adelani et al. [2024].

Languages. We use all 72 languages (English plus 71 targets) that are in MBERT's pretraining list, in XLM-R's, in FLORES-200 and in WIKIANN. Restricting to languages that both models saw removes the unseen-language confound.

Training. We use standard recipes with no per-language tuning: AdamW, a linear schedule with 10% warmup, mixed precision, and model selection on English dev data only. XNLI uses 3 seeds, NER 3 seeds and SIB-200 5 seeds, for 22 runs in total. Section A gives all hyperparameters.

3.3 Predictors

Fertility ratio (primary). We count subword tokens per whitespace word on the language's 1,004 FLORES-200 sentences and divide by the same quantity for English (1.32 for MBERT, 1.40 for XLM-R). Chinese, Japanese, Thai and Burmese are not whitespace-segmented, so we exclude them from the primary analysis. This leaves 67 targets (12 for XNLI).

Parallel token ratio (secondary). We divide the number of tokens on the FLORES-200 sentences by the number of tokens on their English translations. This is defined for all 71 targets.

Baselines. The main competitor is URIEL Littell et al. [2017] syntactic distance from English, the typological distance used most in prior work. We also report phonological, inventory, genetic and geographic distances, and all five jointly with adjusted R^2. Pretraining size is $\log_{10}$ bytes of

the language's Wikipedia dump for MBERT and of its CC-100 file for XLM-R. We also include a non-Latin script indicator.

3.4 Statistical Analysis

We report OLS R^2 with 95% percentile bootstrap confidence intervals over languages (10,000 resamples), and a bootstrap interval for differences in R^2. For H4 we report partial correlations with OLS t-tests and the unique R^2 of each predictor in the model with fertility, log size and syntactic distance. We apply Holm correction across the six cells for each family of tests. For a pooled estimate we z-score the drop within each cell and cluster the bootstrap by language. For H3 we compare the share of languages with $\Delta > 30$ above and below $2.5\times$ with Fisher's exact test, and we add a hinge at 2.5 to the linear fit as a check for nonlinearity.

3.5 Two Designs That Hold the Language Fixed

Across languages, fertility is correlated with size, typology and script. We add two designs in which these cannot vary.

Within-language contrast. For each language we relate the fertility difference between the two models (MBERT minus XLM-R) to the drop difference (MBERT minus XLM-R). Typology and test data are identical on both sides.

Over-segmentation intervention. We re-segment English test inputs so that no subword exceeds a cap of 6, 4, 3, 2 or 1 characters. The text, the model and the labels are unchanged; only the granularity of the segmentation changes. With no cap, our manual pipeline reproduces the standard tokenization exactly; we checked this on 300 examples per task and model and asserted it against the standard English score for every checkpoint. This is the only experiment here that manipulates fertility directly.

3.6 Deviations from the Plan

We trained XNLI on a fixed 100k subset of MNLI instead of all 393k examples because of limited compute, so English accuracy is about 3 points below published full-data numbers. We measure MBERT's pretraining size by Wikipedia dump bytes instead of article counts. We added the script control in H4 after inspecting the fertility values but before running any regression. Section B lists all deviations.

4 Results

4.1 Transfer Performance

Table 2 summarizes zero-shot transfer. Mean drops range from 7.2 points (SIB-200, XLM-R) to 19.9 points (NER, MBERT), with large spread across languages; this spread is what the predictors must explain. The seed standard deviation of a language's score averages 0.4 points (XNLI) to 1.7 points (NER), small relative to the between-language spread. On whitespace-segmented targets, fertility ratios range from 0.92 to 3.80 for MBERT (median 1.64) and from 0.84 to 2.02 for XLM-R (median 1.31). XLM-R's larger vocabulary fragments every language less than MBERT's.

4.2 H1 and H2: Variance Explained

H1 is not supported. Table 3 and figure 1 show that no fertility point estimate falls in the 40–60% range; the values are 0.00–0.27. The upper confidence bound reaches 0.40 only where the interval is widest: the two 12-language XNLI cells (0.60 and 0.40) and SIB-200 with MBERT (0.46). The association is nonetheless real in the larger cells. On NER and SIB-200, Pearson r is 0.33–0.52 and Spearman ρ is 0.40–0.66, all with Holm-adjusted $p < 0.02$. On XNLI it is not distinguishable from zero ($r = 0.40$ and -0.03, $n = 12$). Figure 2 shows the underlying scatter.

H2 is not supported; the sign is reversed. Fertility R^2 is below syntactic-distance R^2 in all six cells. The bootstrap interval of the difference excludes zero in four cells (both XNLI and both NER cells, with differences of -0.24 to -0.64); after Holm correction only NER with XLM-R stays

Task	Model	Seeds	English	Mean target	Mean drop	SD of drop	Largest drop
XNLI	MBERT	3	78.1	64.3	13.8	6.7	27.4 (sw)
	XLM-R	3	80.9	70.8	10.0	4.0	17.2 (sw)
WIKIANN NER	MBERT	3	84.4	64.6	19.9	15.2	83.8 (th)
	XLM-R	3	82.8	64.6	18.2	15.5	79.0 (th)
SIB-200	MBERT	5	88.5	76.5	12.1	9.0	36.5 (mg)
	XLM-R	5	89.5	82.4	7.2	5.5	27.0 (ga)

Table 2: Zero-shot transfer performance over all target languages (14 for XNLI, 71 for NER and SIB-200). Scores are accuracy (XNLI, SIB-200) or span F1 (NER), averaged over seeds; drops are in points relative to English. Uniform chance is 33% for XNLI and 14% for SIB-200.

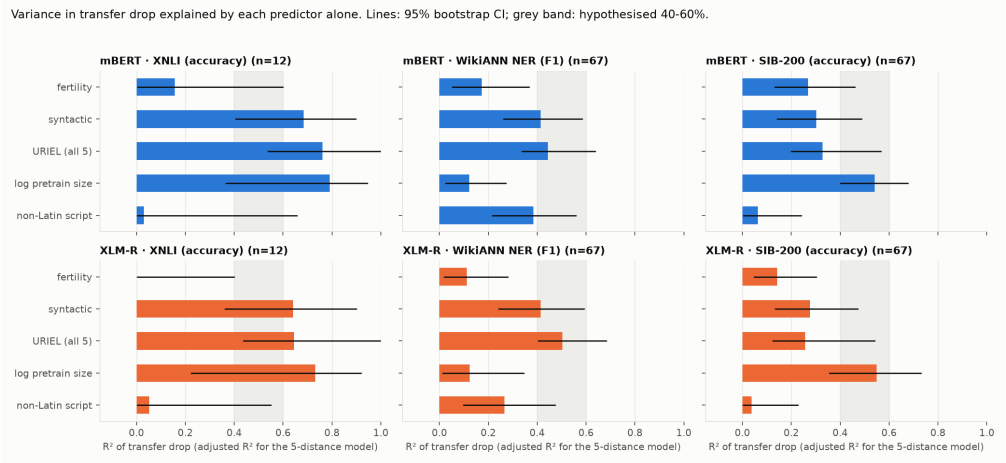


Figure 1: R^2 of transfer drop for each predictor alone, by model (rows) and task (columns). Lines show 95% bootstrap confidence intervals and the grey band marks the hypothesized 40–60% range. The fertility bar falls short of the band in every panel, while syntactic distance or log pretraining size reaches or exceeds it in most.

below 0.05. On SIB-200 the two predictors are statistically indistinguishable (differences of -0.03 and -0.13). Which baseline is strongest depends on the task: log pretraining size is the best single predictor on XNLI and SIB-200, and syntactic distance on NER, where size and fertility are tied at low values.

Pooled estimate. Over 292 language–cell rows from 67 languages, fertility explains 0.16 [0.07, 0.28], syntactic distance 0.37 [0.26, 0.48] and log size 0.32 [0.19, 0.46]. The difference between fertility and syntactic distance is -0.21 [-0.33, -0.08].

4.3 H3: The 2.5× Threshold

Table 4 tests the threshold claim. Under XLM-R's tokenizer no whitespace-segmented language exceeds 2.02×, so we cannot test the claim there; yet 10 languages still lose more than 30 F1 on NER. Under MBERT, six languages exceed 2.5×. NER fits the claim (5 of 6 fail above the threshold against 8 of 61 below; Fisher $p = 0.001$), but topic classification does not (1 of 6 against 3 of 61; $p = 0.32$), and the mean SIB-200 drop above the threshold is 19.7 points.

We find no break at 2.5. Adding a hinge at 2.5 to the linear fit does not help, and its coefficient is negative (NER $p = 0.33$, SIB-200 $p = 0.07$); if anything, the slope flattens.

The NER result is also entangled with script. All six languages above 2.5× use Brahmic or Georgian scripts. In a post hoc check, 6 of the 21 non-Latin targets with fertility *below* 2.5 also lose more than 30 F1, including the four largest drops in the cell (pa, ur, fa, ar, at fertility 1.3–1.9), against 2 of 40 Latin-script targets. With the parallel token ratio over all 71 targets, three languages exceed 2.5 for MBERT (ml, my, th); all three fail on NER and none on SIB-200. We conclude that H3 is not supported as a general rule; the data are consistent with it in one cell of six.

			Single predictor			Joint model	
Task	Model	n	Fertility	Syntactic distance	log pretrain size	URIEL, all 5 (adj.)	Non-Latin script
XNLI	MBERT	12	0.16 [0.00, 0.60]	0.68 [0.40, 0.90]	**0.79** [0.36, 0.95]	0.76	0.03
	XLM-R	12	0.00 [0.00, 0.40]	0.64 [0.36, 0.90]	**0.73** [0.22, 0.92]	0.65	0.05
NER	MBERT	67	0.17 [0.05, 0.37]	**0.41** [0.26, 0.59]	0.12 [0.02, 0.27]	0.44	0.38
	XLM-R	67	0.11 [0.02, 0.28]	**0.41** [0.24, 0.59]	0.12 [0.01, 0.35]	0.50	0.27
SIB-200	MBERT	67	0.27 [0.13, 0.46]	0.30 [0.14, 0.49]	**0.54** [0.40, 0.68]	0.33	0.06
	XLM-R	67	0.14 [0.05, 0.30]	0.28 [0.13, 0.47]	**0.55** [0.35, 0.73]	0.26	0.04
Pooled (292 rows)		67	0.16 [0.07, 0.28]	**0.37** [0.26, 0.48]	0.32 [0.19, 0.46]	–	–

Table 3: R^2 of transfer drop for each predictor alone, with 95% bootstrap confidence intervals in brackets. Fertility never reaches the hypothesized 40–60% and is below syntactic distance in all six cells. The best of the three main single predictors is in **bold**. The joint URIEL column reports adjusted R^2 for a model with five distances. The pooled row z-scores drops within cell and clusters the bootstrap by language.

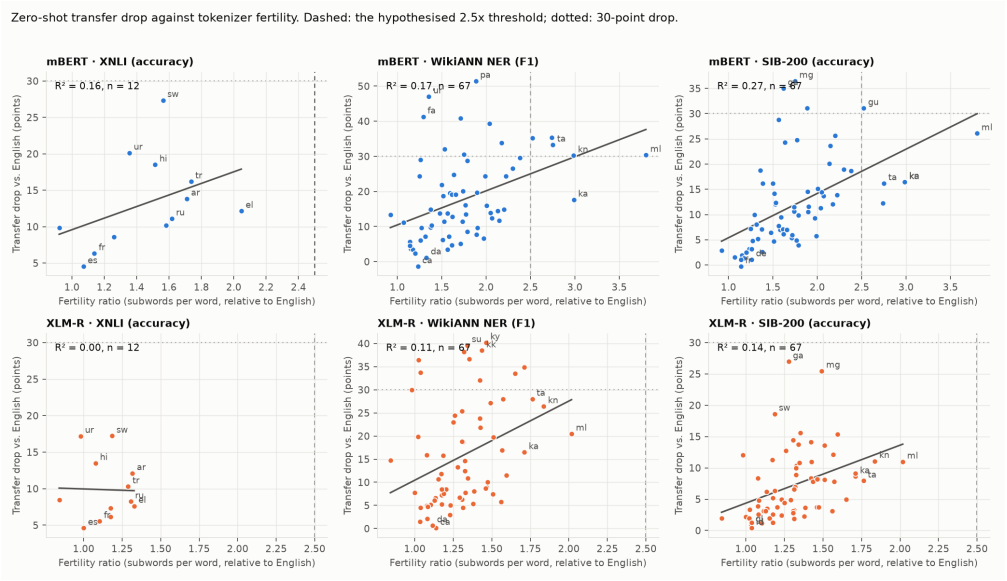


Figure 2: Transfer drop against fertility ratio for each language, with the OLS fit. The dashed line marks the hypothesized 2.5× threshold and the dotted line a 30-point drop. Many languages with low fertility have large drops (e.g., `ur`, `fa`, `sw`, `mg`), and no language exceeds 2.5× under XLM-R (bottom row).

4.4 H4: Independence from Pretraining Size

Table 5 shows that fertility and pretraining size are only moderately correlated ($r \approx -0.3$), so the fertility association is not simply a size effect. Given size alone, the partial correlation stays positive in five cells and is significant in two. Given size *and* syntactic distance, it is significant in one cell (SIB-200 with MBERT), and fertility's unique contribution is at most 0.05 of R^2 everywhere. We find no fertility × size interaction in any cell (all $p > 0.1$). H4 is therefore supported only in a narrow sense: size alone does not explain the association away, but fertility carries little information beyond size and syntax together.

4.5 Within-Language Contrast

For a given language, the more MBERT fragments it relative to XLM-R, the larger MBERT's extra drop (figure 3, table 6). The correlation is 0.61 on XNLI, 0.34 on NER and 0.57 on SIB-200, and it barely changes when we control for the difference in pretraining size. Typology is fixed by construction, so this is the strongest observational evidence here for a fertility effect. It does

Cell	Languages above 2.5×	Above: drop > 30	Below: drop > 30	Mean drop above / below	Fisher p
NER, MBERT	6 (gu ka kn ml ta te)	5 of 6	8 of 61	30.4 / 16.5	0.001
SIB-200, MBERT	same 6	1 of 6	3 of 61	19.7 / 11.2	0.32
XNLI, MBERT	0 of 12	–	0 of 12	–	–
NER, XLM-R	0 of 67	–	10 of 67	–	–
SIB-200, XLM-R	0 of 67	–	0 of 67	–	–
XNLI, XLM-R	0 of 12	–	0 of 12	–	–

Table 4: The 2.5× threshold on whitespace-segmented targets. Only MBERT's tokenizer produces any language above the threshold, and those six languages fail on NER but not on topic classification.

			Partial r of fertility with drop, given			Unique R^2
Task	Model	r(fert., size)	size	size + syntactic	size + syntactic + script	fert. / size / syntactic
XNLI	MBERT	−0.29	0.32 (0.68)	0.38 (1.0)	0.26 (1.0)	0.02 / 0.12 / 0.05
	XLM-R	0.00	−0.05 (0.89)	−0.03 (1.0)	−0.24 (1.0)	0.00 / 0.16 / 0.07
NER	MBERT	−0.33	**0.34** (0.026)	0.09 (1.0)	−0.22 (0.38)	0.005 / 0.001 / 0.20
	XLM-R	−0.33	0.25 (0.18)	0.02 (1.0)	−0.11 (1.0)	0.00 / 0.005 / 0.24
SIB-200	MBERT	−0.33	**0.43** (0.002)	**0.34** (0.031)	0.28 (0.16)	0.05 / 0.27 / 0.01
	XLM-R	−0.33	0.21 (0.27)	0.09 (1.0)	0.08 (1.0)	0.004 / 0.31 / 0.03

Table 5: Does fertility carry information beyond pretraining size? Holm-adjusted p-values in parentheses; partial correlations with adjusted $p < 0.05$ in **bold**. Unique R^2 is the loss in R^2 when a predictor is removed from the model with fertility, log size and syntactic distance. Fertility's unique contribution never exceeds 0.05.

not isolate segmentation, however: the two models also differ in pretraining corpus, objective and training budget.

4.6 Intervention: Over-Segmenting English

Finer segmentation of the same English text causes a monotonic, steep loss in every cell (table 7, figure 4). A 4-character cap, which raises fertility to 1.2–1.5×, costs 10–24 points; a 2-character cap, at 1.8–2.3×, costs 30–59 points. At equal fertility the loss can be much larger than the cross-lingual regression line predicts. On SIB-200 with MBERT at 1.73×, over-segmented English loses 42 points, against about 12 for a typical language at that fertility. Real high-fertility languages do much better than over-segmented English.

What does the intervention measure? It shows that segmentation granularity has a causal effect on these models, with a caveat about what kind. The forced pieces are sequences the model rarely or never saw for those words in pretraining or fine-tuning, so the intervention measures the cost of *unfamiliar* fragmentation. A language that is naturally fragmented was pretrained in that form. The gap between the intervention curve and the cross-lingual points suggests that pretraining exposure absorbs most of the cost, which fits fertility being a weak cross-lingual predictor. This reading is an interpretation; no experiment here tests it directly. At the 1-character cap the two sentence-level tasks also mix fragmentation with truncation; at 2 characters and above, truncation is under 1%.

4.7 Robustness and Exploratory Results

Table 8 reports R^2 under alternative definitions and subsets. The primary conclusion does not depend on linear versus log fertility, or on the ten NER test sets with only 100 sentences.

The fertility definition matters on SIB-200. The parallel token ratio explains 42–51% of the variance there, inside the hypothesized range. It keeps a partial correlation of 0.45–0.60 after controlling for size, syntactic distance and script (uncorrected $p < 0.001$). This is not due to the four added unsegmented languages, since it holds on the same 67 targets. On NER and XNLI the same measure does no better than the primary one, and pretraining size still matches or exceeds it on SIB-200 (0.55–0.56). The SIB-200 test sentences are themselves FLORES-200 sentences, so here the parallel ratio is measured on the same text distribution as the task, which may be part of why it works. The measure was pre-specified as secondary, but its decomposition is post hoc and it is one of several variants, so we treat it as a lead and not as a confirmation.

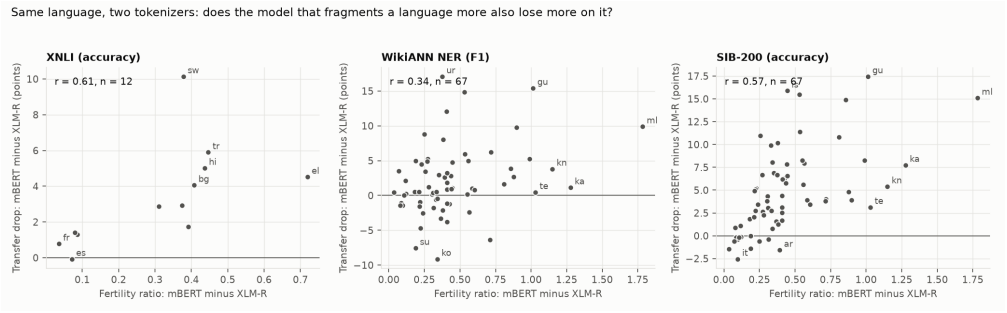


Figure 3: Within-language contrast. Each point is one language; the x-axis is how much more MBERT fragments it than XLM-R, and the y-axis is how much more MBERT loses on it. The relationship is positive on all three tasks ($r = 0.61$, 0.34, 0.57).

Task	n	Pearson r [95% CI]	Spearman ρ	Partial r given size difference
XNLI	12	0.61 [0.30, 0.93]	0.83	0.59 ($p = 0.056$)
NER	67	0.34 [0.13, 0.53]	0.36	0.34 ($p = 0.006$)
SIB-200	67	0.57 [0.38, 0.71]	0.64	0.57 ($p < 0.001$)

Table 6: Within-language contrast: correlation between the fertility difference (MBERT minus XLM-R) and the drop difference (MBERT minus XLM-R) across languages. The language, its typology and its test data are the same on both sides of each difference.

In-domain fertility is a better measurement. Measuring fertility on WIKIANN's own words roughly doubles R^2 on NER (0.36 and 0.28), still below syntactic distance (0.41).

On NER, fertility is a between-script difference. Within either script group, fertility explains essentially nothing on NER (0.00–0.04), so its cell-level R^2 of 0.11–0.17 reflects the difference between Latin and non-Latin targets.

Measurement noise. With 204 test sentences per language, binomial sampling accounts for 17% (MBERT) and 38% (XLM-R) of the between-language variance in SIB-200 drops, which caps any predictor's R^2 near 0.83 and 0.62. Pretraining size (0.54, 0.55) is close to that ceiling for XLM-R; fertility (0.27, 0.14) is not. XNLI drops are 96–98% reliable.

4.8 Error Analysis: Where Fertility Mispredicts

We inspect the residuals of the fertility regression and find three patterns.

Low fertility, large drop. Urdu, Persian, Arabic, Swahili, Malagasy and Irish lose far more than fertility predicts: on NER with MBERT, the residuals are $+33$ for Urdu, $+32$ for Punjabi, $+28$ for Persian and $+23$ for Arabic, and on SIB-200 $+24$ for both Malagasy and Irish. These languages are tokenized compactly but are syntactically distant, low-resource, or (for NER) written in Arabic script. XLM-R tokenizes Urdu at 0.98× English and still loses 17 points on XNLI.

High fertility, small drop. Finnish, Hungarian, Korean and Telugu lose less than fertility predicts. These agglutinative, well-resourced languages are fragmented heavily and transfer fine.

Task dependence. NER residuals split by script (mean about $+5$ for non-Latin and -4 for Latin targets), whereas sentence-classification residuals do not (within 1 point).

5 Discussion

Fertility is a moderate correlate, not the dominant driver. The hypothesis treats fertility as the main, tokenizer-specific cause of transfer loss. Our data describe it as a moderate correlate that is largely redundant with better predictors. Which predictor matters most depends on the task: syntactic distance for NER, pretraining volume for topic classification, and both for NLI. This agrees

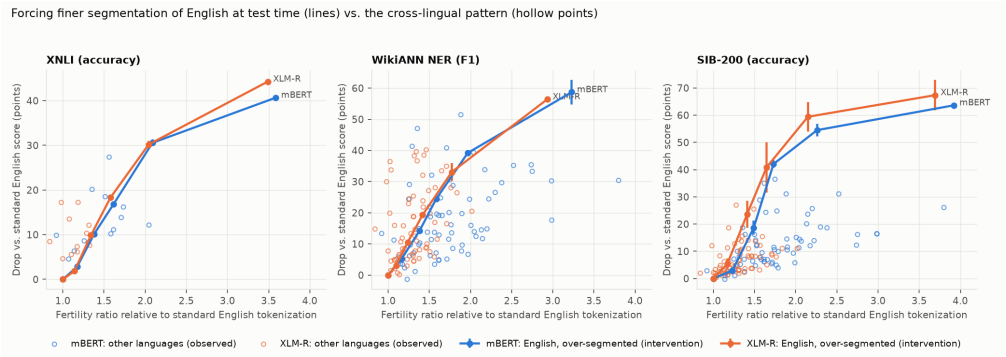


Figure 4: Drop in English score when we force finer segmentation at test time (lines, with seed SD), compared with the observed drops of other languages at their natural fertility (hollow points). The loss grows monotonically and steeply, and on SIB-200 over-segmented English does far worse than natural languages at the same fertility.

		XNLI		NER		SIB-200	
Max subword length	Fertility ratio	mBERT	XLM-R	mBERT	XLM-R	mBERT	XLM-R
6 chars	1.1–1.2	2.8	1.8	5.2	3.0	2.9	5.4
4 chars	1.2–1.5	10.2	9.9	14.3	10.6	18.6	23.6
3 chars	1.4–1.7	16.8	18.3	24.4	19.3	42.1	40.9
2 chars	1.8–2.3	30.6	30.2	39.2	33.1	54.5	59.4
1 char	2.9–3.9	40.7	44.3	58.7	56.4	63.6	67.4

Table 7: Drop in English test score (points) when we cap the length of every subword at test time. The fertility ratio is relative to standard English tokenization and varies by model and task. Values are means over seeds; seed SD is under 1 point for XNLI, up to 4 for NER and up to 9 for SIB-200. At the 1-character cap, 8% (SIB-200) and 18–19% (XNLI) of tokens are also truncated by the length limit.

with the typology-and-size account of Lauscher et al. [2020] and with the findings of Wu and Dredze [2020] on low-resource languages.

Fertility is not irrelevant either. The within-language contrast and the intervention both point to a real effect of fragmentation. The reconciliation we find most plausible is that fertility mixes two things: fragmentation that the model was never trained to handle, which is harmful, and fragmentation that reflects ordinary morphology in a well-resourced language, which is mostly harmless. A single ratio cannot tell them apart, which would explain why Finnish and Malayalam can both be heavily fragmented and have very different outcomes. This is a hypothesis generated by our results, not something they establish.

Surprises. First, XLM-R never reaches the hypothesized danger zone for any whitespace-segmented language, so a threshold defined on fertility is relative to a tokenizer and does not carry over to another. Second, the choice of fertility definition changes the SIB-200 answer from weak to moderately strong. Third, over-segmented English does far worse than natural languages at the same fertility.

Practical implications. Fertility alone is a weak screening signal for transfer risk. Syntactic distance and pretraining volume are more informative, and for NER so is script. Practitioners should not use the 2.5× threshold as a rule.

Limitations.

- *Two models.* Both are base-size encoders. Our results do not speak to decoder-only language models or to byte- and character-level models.
- *Observational design.* Fertility, size, typology and script are correlated across languages, and with 67 languages (12 for XNLI) partial effects have wide intervals. The XNLI cells cannot rule out a fertility R^2 of 0.4–0.6 (upper bounds 0.60 and 0.40), although the point estimates are 0.16 and 0.00.

Variant	XNLI		NER		SIB-200	
	MBERT	XLM-R	MBERT	XLM-R	MBERT	XLM-R
Primary (subwords per word, linear)	0.16	0.00	0.17	0.11	0.27	0.14
log fertility	0.19	0.00	0.18	0.10	0.31	0.15
Parallel token ratio, all targets	0.33	0.01	0.27	0.00	**0.51**	**0.42**
Parallel token ratio, segmented targets (post hoc)	0.14	0.00	0.22	0.03	**0.50**	**0.44**
Fertility measured on WIKIANN words	–	–	0.36	0.28	–	–
NER test sets of 1,000+ sentences	–	–	0.16	0.09	–	–
Latin-script targets only	0.52†	0.15†	0.04	0.03	0.32	0.18
Non-Latin targets only	0.34†	0.49†	0.00	0.01	0.23	0.07

Table 8: R^2 of transfer drop on fertility under alternative definitions and subsets. Values inside the hypothesized 40–60% range with $n = 67$ or more are in **bold**. † $n = 6$ languages.

- *Proxies.* MBERT's pretraining size uses 2023 Wikipedia dump sizes, not the 2018 dump the model was trained on, and XLM-R's uses the public CC-100 re-creation. We computed URIEL distances ourselves from kNN-imputed features. We compiled XLM-R's language list from our recollection of the original paper and excluded Haitian and Yoruba as uncertain.
- *Reduced* XNLI *training.* Training on 100k examples lowers absolute accuracy; we did not check whether it changes relative drops.
- *Noisy cells.* SIB-200 has 204 test sentences per language and ten WIKIANN languages have 100. WIKIANN labels are automatically derived and uneven in quality, which adds language-specific noise unrelated to any predictor.
- *Whitespace fertility.* It is undefined for four languages and is a crude proxy for words in others; the sensitivity to the fertility definition in section 4 shows that this matters.
- *Intervention validity.* The intervention creates out-of-distribution segmentations, so it measures the cost of unfamiliar fragmentation. It does not reproduce what a naturally high-fertility language experiences.
- *One recipe.* We use one learning rate per task and no per-language tuning. Seed variance is small relative to between-language variance, but we did not explore recipe sensitivity.
- *Literature coverage.* We did not conduct a systematic literature review, so our positioning relative to prior work may be incomplete.

6 Conclusion

We tested whether tokenizer fertility explains zero-shot cross-lingual transfer loss in MBERT and XLM-R across three tasks and up to 71 target languages. Fertility explains 0–27% of the between-language variance in transfer drop, not 40–60%. Syntactic distance outperforms it in every setting we tested, the 2.5× threshold is not a reliable failure boundary, and fertility's predictive value beyond pretraining size and syntactic distance is small. Fragmentation does have a causal cost when it is unfamiliar to the model, and a tokens-per-parallel-sentence measure shows a stronger association that survives controls on one task. The key takeaway is that how finely a tokenizer splits a language tells us little, by itself, about how well a model will transfer to it.

Several follow-ups would sharpen these conclusions. The most valuable is to train the same architecture on the same corpus with different vocabularies; this is the experiment that isolates the tokenizer, and nothing here does. Testing the parallel token ratio on other parallel benchmarks would show whether the SIB-200 result is general or an artifact of measuring fertility on the task's own text distribution. Replacing raw fertility with a measure of *excess* fragmentation, for example relative to a monolingual tokenizer Rust et al. [2021] or to morpheme counts, would separate the two components we propose in section 5. Finally, the analysis should be repeated with full-data XNLI and with larger and decoder-only models.

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A Training Details

All runs use AdamW with a linear learning-rate schedule and 10% warmup, mixed precision, an evaluation batch size of 128, and model selection on English dev data only. Table 9 lists the task-specific settings. NER uses first-subword labeling and span F1 computed with seqeval. For SIB-200 we keep the checkpoint from the best English-dev epoch.

Task	Epochs	Batch size	Learning rate	Max length	Seeds
XNLI	2	32	2×10^{-5}	128	3
NER	5	32	3×10^{-5}	256	3
SIB-200	20	16	2×10^{-5}	–	5

Table 9: Fine-tuning hyperparameters. The same settings are used for both models.

Environment. We ran all experiments on four NVIDIA RTX A6000 GPUs (48 GB) shared with unrelated jobs, using Python 3.11.14, torch 2.14.1, transformers 5.19.0 and tokenizers 0.23.3. Wall time per run was 2–4 minutes for SIB-200, 17–56 minutes for NER and 29–45 minutes for XNLI. Bootstrap resampling uses seed 42.

B Deviations from the Pre-Specified Plan

We wrote the analysis plan before training any model and logged the following deviations.

1. XNLI *training set.* We trained on a fixed random 100k subset of MNLI (the same subset for all seeds and both models) instead of all 393k examples, because the shared GPUs ran at 2–4 steps per second. English accuracy is about 3 points below published full-data numbers.
2. MBERT *pretraining size.* We use the byte size of each language's November 2023 Wikipedia dump instead of article counts, because the Wikimedia API rate-limited our requests. Neither is the 2018 dump on which MBERT was trained.
3. URIEL *distances.* The installed lang2vec package ships feature vectors but not precomputed distances, so we computed cosine distances on kNN-imputed feature vectors, and Euclidean distance for the geographic features. Basque has an empty family vector, so we set its genetic distance to 1.
4. *Small NER test sets.* Ten WIKIANN targets have only 100 test sentences. We keep them in the primary analysis because the language set was fixed in advance, and we report a robustness row restricted to test sets of 1,000 or more sentences (table 8).
5. *Script control.* We added script as a control in H4 after seeing that the MBERT languages above 2.5$\times$ use Brahmic or Georgian scripts, but before running any regression.